

Prateek Nigam

Data Scientist

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SUMMARY

A data science professional with 5+ years of experience specializing in Market Mix Modelling, Gen AI multi-agent systems and advanced Language modeling, currently pursuing a Master's in Computer Science with AI. Skilled in implementing analytics models, driving data models with strong mathematical grasp of machine learning concepts.

SKILLS

Technical Skills: Python, Transformers, Fine-Tuning, RAG, LLMs, Multi-Agents, Machine Learning (Regression, Decision Trees), Deep Learning (Pytorch), Bayesian Regression, Statistics, Data Analysis, Data Visualization (MS Excel), MCP Concept for multi agent Designing.

Strategic Problem Solving: Critical and analytical thinking, aligning data insights with business strategies, particularly in MMM, with strong grasping skills to quickly adapt to new domains.

EXPERIENCE

Fractal Analytics

Oct 2021 - Aug 2024

Data Scientist

Mumbai, India

Market Mix Modelling

- Contributed as a **subject matter expert in MMM (Market Mix Modeling)** across CPG/FMCG product categories – analyzed marketing effectiveness – **optimized advertisement investments**, leading to a **projected 5-10 % increase in ROI** and profitability.
- Optimized **Hierarchical Bayesian Modeling** run time - **from 5 hours to 8 mins** by replacing theano tensors with Jax arrays - thus enhancing efficiency allowing to handling larger datasets and **increasing project capacity by 50%**.
- Faced the need for insightful and detailed **model at the zip code level** – researched and built technical models using **LGBM modeling** with a performance of around **70%** – helped improve targeted marketing.
- Addressed the **problem of redundant data analysis** – **automated the complete EDA process** – saved approximately **one-third in analysis time**, boosting overall productivity.

Copy Testing and Survey Analysis

- Conducted a copy testing exercise to analyze different datasets, focusing on **KPIs** like **customer retention**, **ad relevance**, and **brand appeal** – **identified effective source cross feature combinations** for predicting **Volume per GRP**.
- Built **regression models for kpi attribution** and created **decision trees** using information gain – **worked closely with the client team** to achieve an **F1 score of about 48%** through **manual tree creation and integration of business insights**.

Capgemini India Pvt Ltd.

Aug 2019 - Oct 2021

Software Engineer

Bangalore, India

- Collaborated** with cross-functional teams to design **solutions** based on evolving client requirements.
- Enhanced the PySpark transformation ETL tool, improving **data processing efficiency** and performance.
- Implemented **Single Sign-On (SSO)** using **Kerberos** and integrated LDAP for Active Directory group filtering.

EDUCATION

University Of Galway, Ireland

2024 – Current

Msc - Computer Science with Artificial Intelligence

Acropolis Institute of Technology and Research

2015 – 2019

B.Tech - Electronics and Communication Engineering

CGPA - 7.4/10

PROJECTS

- Building the Beast - Transformers from Scratch** 🐞 - Built transformer models from scratch to learn how they work. I broke down each part to understand how they fit together.
- AutoMMM Lab** 🐞 - AI-powered Market Mix Modeling system that autonomously analyzes ad spend, sales, and external factors using multi-agent collaboration with shared context.